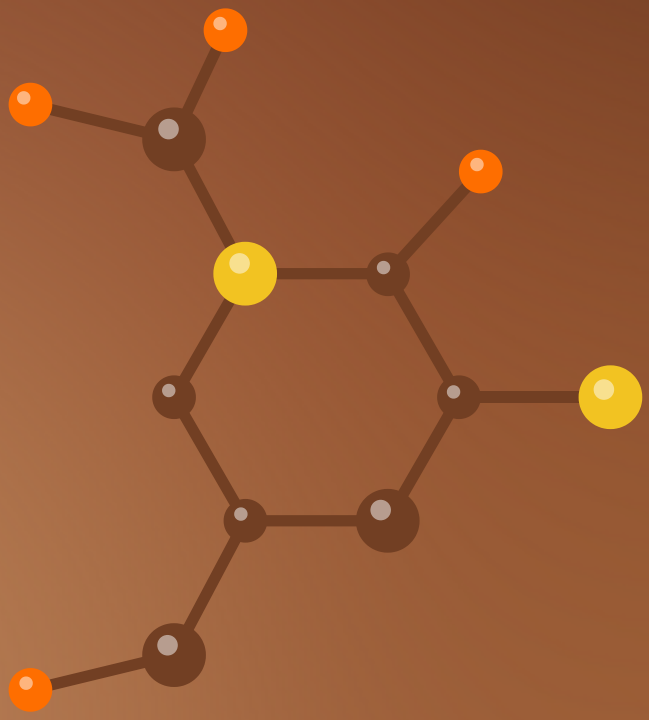
ALLEN CAREER INSTITUTE
path to success
KOTA (RAJASTHAN)



ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Biomolecules

English Medium

EXERCISE # O-I

1. A D-carbohydrate is :

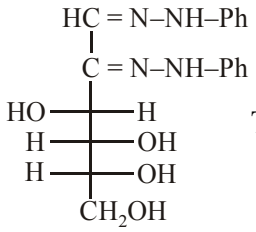
(A) Always dextrorotatory

(B) Always laevorotatrory

(C) Always the mirror of the corresponding L-carbohydrate

(D) None of these

BM0007

2. 

The given osazone can be obtained by :

(A) D-glucose (B) D-mannose (C) D-Idose (D) Both (A) & (B)

BM0008

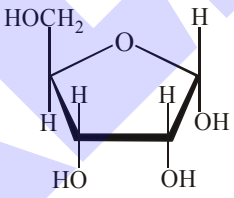
3. Which of the following pair gives same phenyl osazone ?

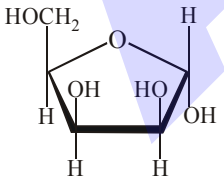
(A) D-Glucose and D-Allose (B) D-Glucose and D-Altrose

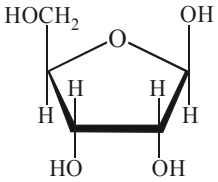
(C) D-Glucose and D-Mannose (D) D-Glucose and D-Talose

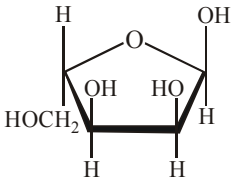
BM0009

4. Which of the following represents the anomer of the compound shown ?



(A) 

(B) 

(C) 

(D) None of these

BM0010

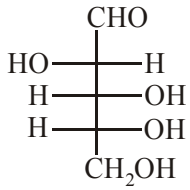
5. For the complex conversion of D-glucose into the corresponding osazone, the minimum number of equivalents of phenyl hydrzine required is :

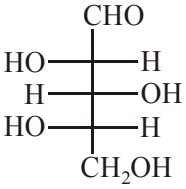
(A) Two (B) Three (C) Four (D) Five

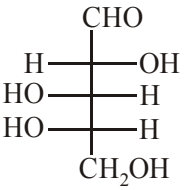
BM0011

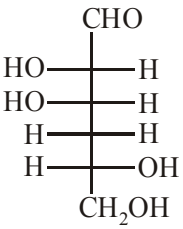
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6. Which of the following structure is L-arabinose ?

(A) 

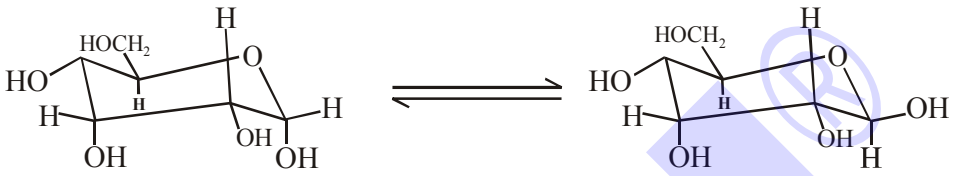
(B) 

(C) 

(D) 

BM0012

7. Which one of the statements concerning the equilibrium shown is true ?



(A) The two structures are enantiomers of each other. They have equal but oppostie optical rotations and recemize slowly at room temperature

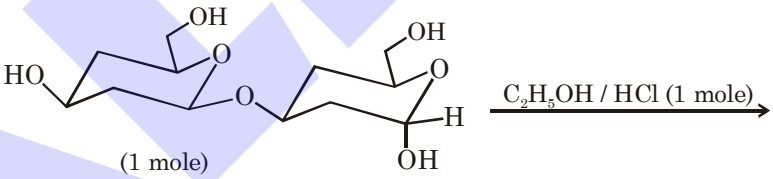
(B) The two structures are enantiomers of each other. They racemize too rapidly at room temperature for their optical rotations to be measured

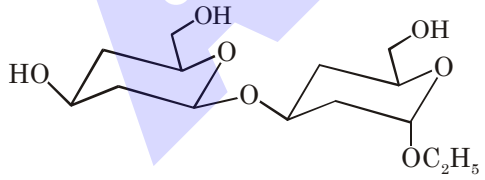
(C) The two structures are diastereomers of each other. Their interconversion is called mutarotation

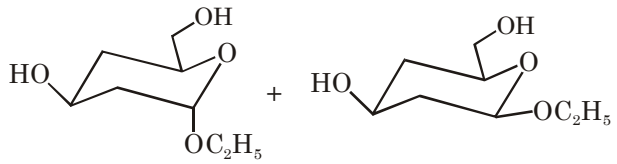
(D) The two structures are diastereomers of each other. Their interconversion does not require breaking and making bonds, only a change in conformation

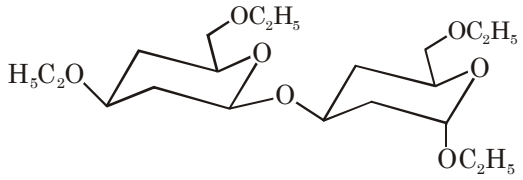
BM0013

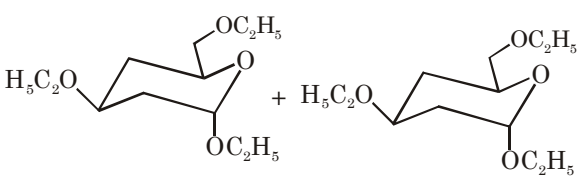
8. Major product of following reaction is :



(A) 

(B) 

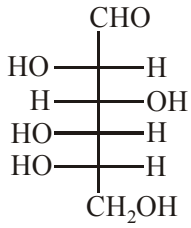
(C) 

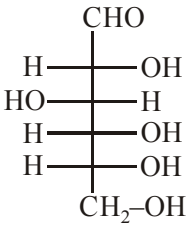
(D) 

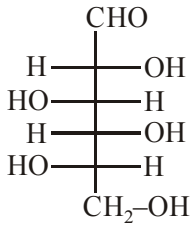
BM0014

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9. What is the structure of L-glucose ?

(A) 

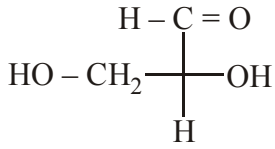
(B) 

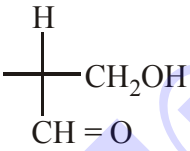
(C) 

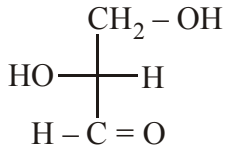
(D) None of these

BM0015

10. What is the structure of L-glyceraldehyde ?

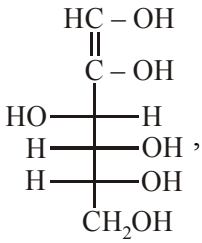
(A) 

(B) 

(C) 

(D) Both (A) and (B)

BM0016

11. 

, the given is enol form of :

(A) D-glucose (B) D-mannose (C) D-fructose (D) All of these

BM0017

12. D-glucose & D-fructose can be differentiated by :

(A) Fehling solution (B) Tollen's reagent (C) Benedict test (D) Br_2 / H_2O

BM0018

13. D-Glucose exist in x different forms. The value of x (stereoisomer) is :

(A) 2 (B) 3 (C) 4 (D) 5

BM0019

14. D-Mannose $\underset{}{\overset{HO^-}{\rightleftharpoons}}$ D-Glucose $\overset{HO^-}{\rightleftharpoons}$ (A) ;

Product (A) of above reaction is :

(A) D-glucose (B) D-fructose (C) D-talose (D) D-idose

BM0020

15. Which of the following statement is not correct for maltose.

(A) It is a disaccharide (B) It undergoes mutarotation

(C) It is a reducing sugar (D) It does not have hemiacetal group.

BM0021

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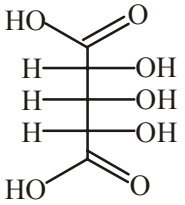
16. Maltose is made up of to two sugars–

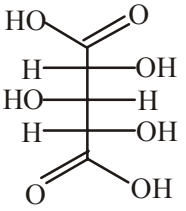
(A) Glucose (B) Fructose

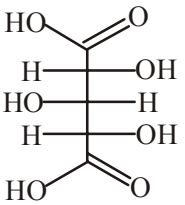
(C) Glucose & galactose (D) Glucose & fructose

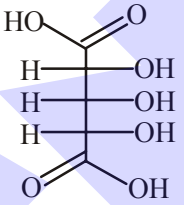
BM0023

17. D-Ribose when treated with dilute HNO_3 forms.

(A)  , Achiral

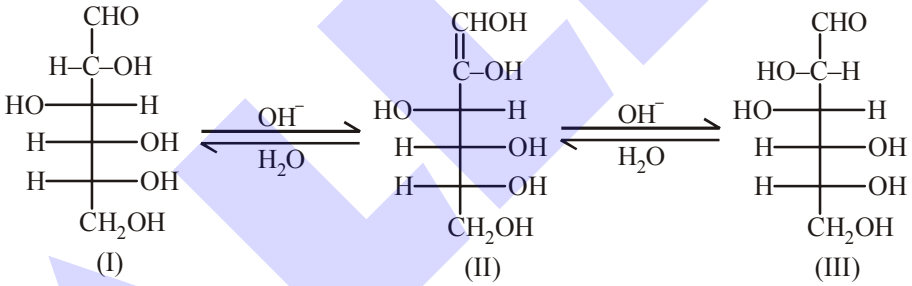
(B)  , Chiral

(C)  , Achiral

(D)  , Chiral

BM0024

18. Consider the given process



and identify the incorrect statement.

(A) Configuration at C–2 is lost on enolisation

(B) I and III are epimers

(C) Proton transfer from water to C–1 converts ene diol to an aldose.

(D) D-glucose can isomerise to D-fructose through enol intermediate.

BM0025

19. When methyl D-glucopyranoside is treated with HIO_4 its number of mole consumed per mole of the sugare is -

(A) 2 (B) 3 (C) 4 (D) 5

BM0026

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20. The configuration of the C-2 epimer of D-glucose is-

(A) 2R, 3S, 4R, 5S
(B) 2S, 3S, 4R, 5R
(C) 2S, 3R, 4S, 5R
(D) 2R, 3S, 4R, 5R

BM0028

21. Mutarotation involve-

(A) Racemisation
(B) Diastereomerisation
(C) Optical resolution
(D) Conformational inversion

BM0029

22. Consider the reaction sequence -

$$\text{Glucose} \xrightarrow[\text{excess}]{PhNHNH_2} P \xrightarrow[\Delta]{H_3O^+} Q \xrightarrow[AcOH]{Zn} R$$

The product R is -

(A) Arbinose (B) Sorbitol (C) Fructose (D) Mannose

BM0030

23. Same osazone derivative is obtained in case of D-glucose, D-Mannose and D-Fructose due to

(A) The same configuration at C-5
(B) The same constitution.
(C) The same constitution at C-1 and C-2
(D) The same constitution and acid configuration at C-3, C-4, C-5 and C-6 but different constitution and configuration at C-1 and C-2 which becomes identical by osazone formation.

BM0033

24. $\text{D(–) –Erythrose} \xrightarrow{NaBH_4} (P)$

$\text{D(–) –Threose} \xrightarrow{NaBH_4} (R)$

Which of the following statement is correct about P and R ?

(A) Both are optically active
(B) Both are optically inactive
(C) P is optically inactive and R is optically active
(D) Neither P nor R has asymmetric carbon.

BM0034

25. A segment (X) of cellulose obtained on partial hydrolysis has molecular mass 1476 gm. On complete acidic hydrolysis, mass of the product obtained is 1620 gm. Find out the number of glycosidic linkage(s) present in segment (X) :

(A) 8 (B) 6 (C) 4 (D) 10

BM0040

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26. Carbohydrates may be :
(A) Sugars
(B) Starch
(C) Polyhydroxy aldehyde/ ketones
(D) Compounds that can be hydrolysed to sugar

BM0043

27. Select the correct statement :
(A) Glycosides do not undergo mutarotation
(B) All OH groups of a cyclic monosaccharides are converted to ethers by treatment with base and an alkyl halide
(C) α-D glucose reacts with Ag_2O and excess CH_3I to form tetramethyl ether
(D) D-glucose upon treatment with warm HNO_3 forms D-glucaric acid

BM0044

28. Starch molecules are polymer with repeating glucose units. Select the correct statement(s).
(A) Glucose units are joined through α-glycosidic linkage
(B) The branches of amylopectin are linked to the chain with α-1,6'-glycosidic linkages
(C) The linear linkages of amylopectin are formed by α-1,6'-glycosidic bond
(D) Amylose has an unbranched skeleton of glucose molecules with α-1,4'-glycosidic linkages

BM0046

29. Which of the following carbohydrates developes blue colour on treatment with iodine solution ?
(A) Glucose (B) Amylose (C) Starch (D) Fructose

BM0049

30. Select the correct statement among following
(A) Number of chiral atom in α-D-glucose is less than D-glucose
(B) D-glucose and D-fructose give same product with HIO_4
(C) D-glucose and D-fructose give same product with H_2NOH
(D) D-glucose and D-fructose form same product with $H_2N–NH–Ph$

BM0055

31. Match the column :

Column I	**Column II**
(A) Sucrose	(P) Two acetals
(B) Maltose	(Q) No hemiacetal
(C) Lactose	(R) β–1,4'-glycosidic bond
(D) Cellulose	(S) One of the hydrolysis product is glucose

BM0058

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EXERCISE # O-II

1. Select the correct option.

(A) Isoelectric point is the pH at which an amino acid exists primarily in its neutral form.

(B) Isoelectric point is the average of pK_a values of α-COOH amino α- NH_3^+ groups [valid only for neutral amino acid]

(C) Glycine is characterised by two pK_a values

(D) For neutral amino acid the concentration of zwitter ion is maximum at its isoelectric point

BM0047

2. Amino acids are synthesised from

(A) α-Halo acids by reaction with NH_3

(B) Aldehydes by reaction with NH_3 and cyanide ion followed by hydrolysis

(C) Pyruvic acid is treated with NH_3 followed by addition of H_2(Ni)

(D) Alcohols by reaction with NH_3 and CN^- ion followed by hydroysis.

BM0048

3. Peptide linkage is -

(A) $-\overset{O}{\overset{\|}{C}}-O-$ (B) $-\overset{O}{\overset{\|}{C}}-NH_2$ (C) $-\overset{O}{\overset{\|}{C}}-NH-$ (D) $-\overset{O}{\overset{\|}{C}}-NH-NH_2$

BM0032

4. The N-base which differenciate DNA with RNA is :

(A) Cytosine (B) Uracil (C) Adenine (D) Guanine

BM0056

5. Iso-electric point of alanine is (pH = 6). At which pH, maximum concentration of zwitter ion of alanine will be present ?

(A) pH > 6 (B) pH < 6 (C) pH = 6 (D) pH = 7

BM0001

6. At iso-electric point :

(A) Concentration of cation is equal to concentration of anion

(B) Net charge is zero

(C) Maximum conc. of di-pole ion (Zwitter ion) will be present

(D) All of the above

BM0002

7. Which of following amino acid has lowest iso-electric point ?

(A) Glycine (B) Alanine (C) Aspartic acid (D) Lysine

BM0003

8. $H-C\equiv C-H \xrightarrow[H_2SO_4]{HgSO_4} (A) \xrightarrow[(2)\ H_3O^+]{(1)\ NH_3+HCN} (B)$; Product (B) of given reaction is :

(A) Glycine (B) Alanine (C) Valine (D) Leucine

BM0004

9. Which amino acid does not contain chiral centre ?

(A) Valine (B) Leucine (C) Glycine (D) Iso-leucine

BM0005

10. Which of the following is Sanger reagent ?

(A) 2,4-Di-nitro flurobenzene (B) Phenyl isocyanane

(C) 2,4-Di-nitro chlororbenzene (D) 12,4-Di-nitro-iodobenzene

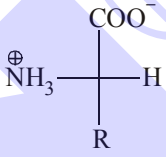
BM0006

11. α - amino acid when heated with NaOH/CaO forms -

(A) α,β - unsaturated acid (B) α,β - unsaturated amine

(C) Carboxylic acid (D) Amine

BM0027

12. The pH of the solution containing following zwitter ion species is 

(A) 4 (B) 5 (C) 7 (D) 9

BM0031

13. If on a strand of DNA the base sequence is ATTGACGCAT then the sequence transcription on RNA would be -

(A)UAACUGCGUA (B) AUUCUGCGUA

(C) UAACTGCGUA (D) TAACTGCGTA

BM0054

14. The monomer of nucleic acids are held together by

(A) Phosphodiester linkage (B) Amide linkage

(C) Glycosidic linkage (D) Ester linkage

BM0035

15. Number of hydrogen bonds between adenine and thymine?

(A) 1 (B) 2 (C) 3 (D) 4 **BM0223**

16. Number of hydrogen bonds between guanine and cytosine?

(A) 1 (B) 2 (C) 3 (D) 4 **BM0224**

17. Which ratio is constant for DNA?

(A) A + G / T + C (B) A + T / G + C (C) A + C / U + G (D) A + U / G + C

BM0225

18. According to Chargaff's rule, in a DNA molecule

(A) The amount of adenine and thymine is equal to the amount of guanine and cytosine

(B) The amount of adenine and guanine is equal to the amount of thymine and cytosine

(C) The amount of adenine and uracil is equal to the amount of guanine and cytosine

(D) The amount of adenine and guanine is equal to the amount of uracil and cytosine

BM0226

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19. The monomeric unit of nucleic acid are called _____

(A) Nucleotides
(B) Nucleosides
(C) Pyrimidines
(D) Purines

BM0227

20. (a) DNA $\xrightarrow{(b)}$ mRNA $\xrightarrow{(c)}$ (d)

(A) (a) Transduction ; (b) Translation ; (c) Replication ; (d) Protein
(B) (a) Replication ; (b) Transcription (c) Transduction ; (d) Protein
(C) (a) Translation ; (b) Replication ; (c) Transcription ; (d) Transduction ;
(D) (a) Replication ; (b) Transcription ; (c) Translation ; (d) Protein

BM0228

21. Which of the following statement is not correct with respect to Deoxyribonucleic acid (DNA)?

(A) It is found in all prokaryotic and eukaryotic cells and in many viruses.
(B) The DNA molecule consists of a single strand that is made of deoxyribose and phosphate groups.
(C) DNA codes genetic information for the trnsmission of interited traits.
(D) Each strand has a backbone made of alternating sugar (deoxyribose) and phosphate groups.

BM0229

22. Which of the following indicates a nucleoside present in RNA?

(A) Adenine
(B) Cytosine
(C) Uridine
(D) Deoxythymidine

BM0230

23. The double helical structure of DNA is developed through ;

(A) Scanning electron microscopy
(B) X-ray crystallography
(C) Compound microscope
(D) Ultracentrifugation

BM0231

24. A segment of DNA has 250 thymine and 250 guanine bases. The total number of nucleotides present in the segment is :

(A) 250
(B) 500
(C) 750
(D) 1000

BM0232

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EXERCISE # S-I

Assertion Reason

1. **Statement 1 :** Furanose ring, like pyranose rings are not planar.
Statement 2 : The most stable conformation of furanose is envelope form.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

BM0060

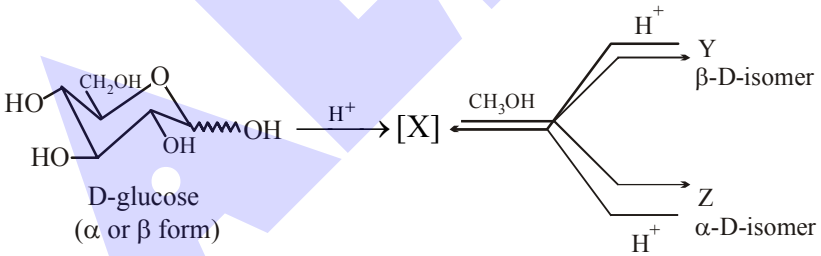
2. **Statement 1 :** Bromine water changes glucose to gluconic acid.
Statement 2 : Bromine water acts as oxidising agent.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

BM0061

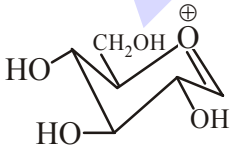
3. **Statement 1 :** All monosaccharide ketoses are reducing sugars.
Statement 2 : Monosaccharide ketose give positive Tollen's and Fehling's test.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **BM0062**

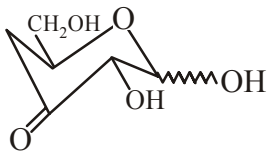
Comprehension (Q.4 to Q.6)

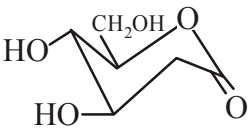
Consider the following reversible process for a reaction of D-glucose.

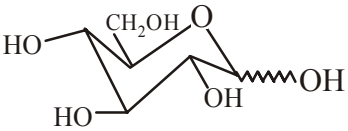


4. The structure of intermediate [X] is

(A) 

(B) 

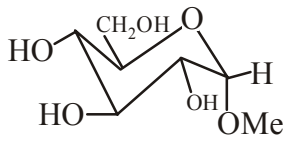
(C) 

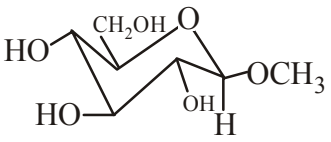
(D) 

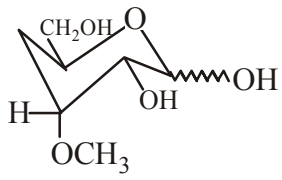
BM0063

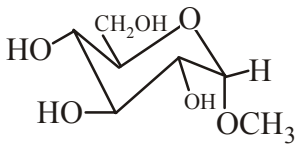
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5. Select the correct option.

(A) Y is 

(B) Z is 

(C) Y is 

(D) Z is 

BM0064

6. Select the correct statement.

(A) Y is more stable than Z due to H-bond

(B) Y is less stable than Z due to anomeric effect.

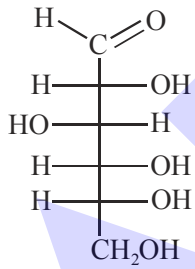
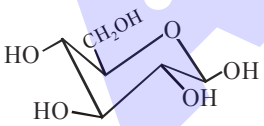
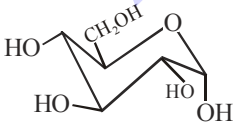
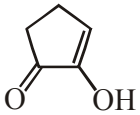
(C) Y is more stable than Z due to anomeric effect

(D) Y is less stable than Z due to H-bond.

BM0065

Match the Column

7. Match the compounds given below existing in equilbrium mixture with their percentage given in the right hand side.

Column I	**Column II**
(A) 	(P) 36 %
(B) 	(Q) 63.8 %
(C) 	(R) 99.9 %
(D) 	(S) 0.2 %

BM0069

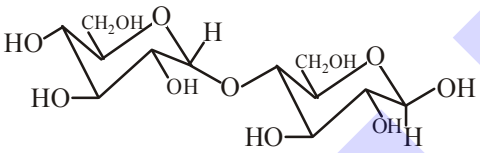
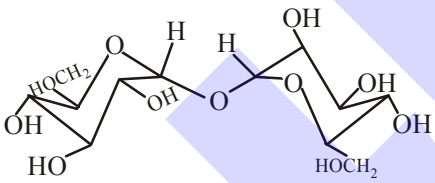
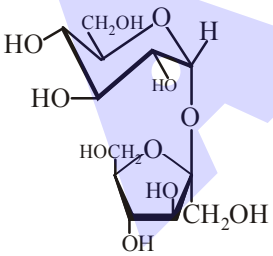
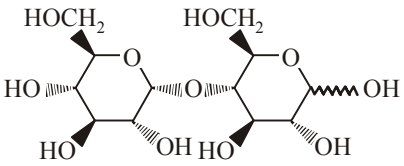
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8. Match the column :

Column I (Carbohydrate)	**Column II (Properties)**
(A) Starch	(P) Mutarotation
(B) Sucrose	(Q) Non reducing sugar
(C) Lactose	(R) β-glycosidic bond
(D) Maltose	(S) α-glycosidic bond
	(T) Reducing sugar
	(U) Hemiacetal

BM0070

9. Match the column :

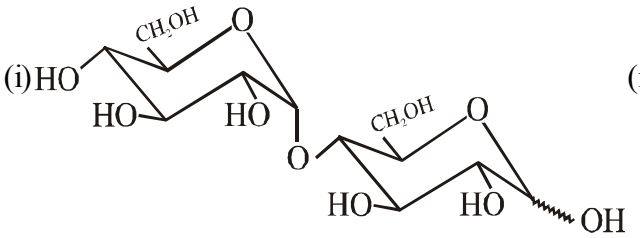
Column I	**Column II**
(A) 	(P) α-glycoside bonds
(B) 	(Q) Reducing sugar
(C) 	(R) Forms enediol intermediate
(D) 	(S) β-glycoside bond

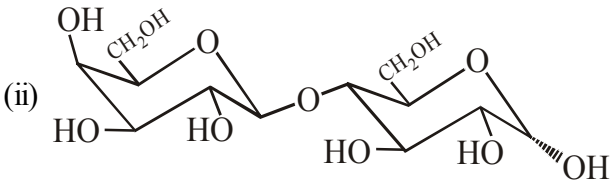
BM0071

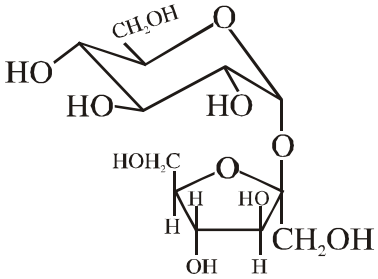
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Subjective Type :

10. For given compound

(i) 

(ii) 

(iii) 

(a) Which glycosidic linkage is present

(b) Name of the compound

(c) Products obtained on hydrolysis

(d) Is it reducing sugar

(e) Number of moles of HIO_4 consumed

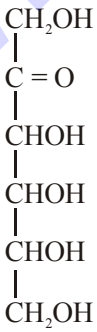
(f) Is mutarotation possible

(g) Number of moles of CH_3COCl consumed per mole of compound.

BM0107

11. For given compound

(a) 

(b) 

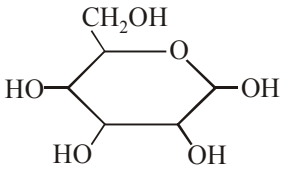
(i) Number of possible isomers

(ii) Number of possible D-configuration isomers

(iii) Number of possible L-configuration isomers

BM0108

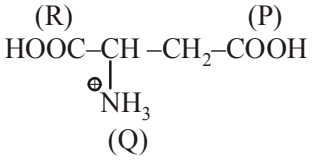
12. Number of possible isomers of α-D-gluco pyronose configuration.



BM0109

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13. The pKa values for the three acidic group P,Q,R are 4.3, 9.7 and 2.2 respectively



Calculate the isoelectric point of the amino acid ?

BM0075

14. Account for the fact that 2-amino ethanoic acid (glycine) exist as a dipolar ion as does p-aminobenzene sulphonic acid but p-amino benzoic acid does not.

BM0106

15. Calculate isoelectric point of following amino acid

	pKa_1 (α-COOH)	pKa_2 (α-COOH)	pKa (side chain)
(a) Glycine $\overset{\oplus}{N}H_3 - CH_2 - COOH$	2.3	9.6	—
(b) Serine $CH_2(OH) - CH(\overset{\oplus}{N}H_3) - COOH$	2.2	9.2	—
(c) Alanive $CH_3 - CH(\overset{\oplus}{N}H_3) - COOH$	2.3	9.7	—
(d) Asparric acid $HOOC - CH_2 - CH(\overset{\oplus}{N}H_3) - COOH$	2.1	9.8	3.9
(e) Lysine $\overset{\oplus}{N}H_3–CH_2–CH_2–CH_2–CH_2–CH(\overset{\oplus}{N}H_3)–COOH$	2.2	9.0	10.5

BM0110

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EXERCISE # J-MAIN

1. Which of the following pairs give positive Tollen's test? **[J-Main-2004]**

(1) Glucose, sucrose (2) Glucose, fructose

(3) Hexanal, Acetophenone (4) Fructose, sucrose

BM0111

2. Two forms of D – glucopyranose, are called. **[J-Main-2005]**

(1) Enantiomers (2) Anomers (3) Epimers (4) Diastereomers

BM0112

3. **Statement-1 :** Glucose gives a reddish-brown precipitate with Fehling's solution.

because **[J-Main-2007]**

Statement-2 : Reaction of glucose with Fehling's solution gives CuO and gluconic acid.

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(3) Statement-1 is True, Statement-2 is False.

(4) Statement-1 is False, Statement-2 is True.

BM0114

4. Which one of the following bases is **not present** in DNA ? **[J-Main-2014]**

(1) Cytosine (2) Thymine (3) Quinoline (4) Adenine

BM0115

5. Which of the vitamins given below is water soluble? **[J-Main- 2015]**

(1) Vitamin E (2) Vitamin K (3) Vitamin C (4) Vitamin D

BM0117

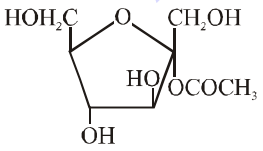
6. Thiol group is present in : **[J-Main-2016]**

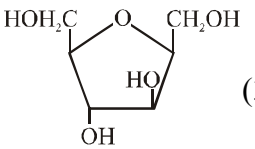
(1) Methionine (2) Cytosine (3) Cystine (4) Cysteine

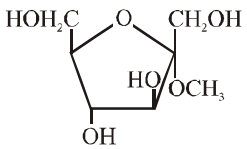
BM0118

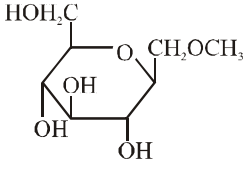
7. Which of the following compounds will behave as a reducing sugar in an aqueous KOH solution

[J-Main-2017]

(1) 

(2) 

(3) 

(4) 

BM0121

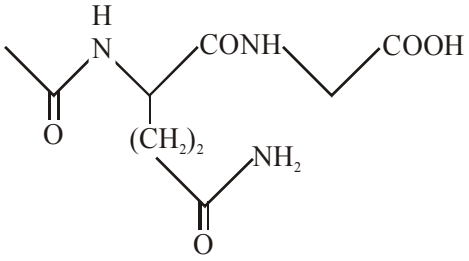
8. Glucose on prolonged heating with HI gives : **[J-Main-2018]**

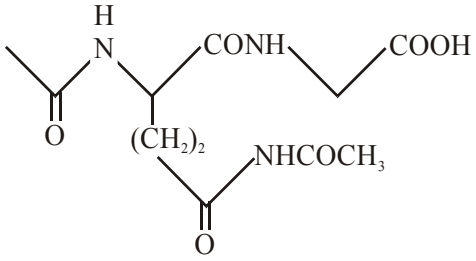
(1) 1–Hexene (2) Hexanoic acid (3) 6-iodohexanal (4) n-Hexane

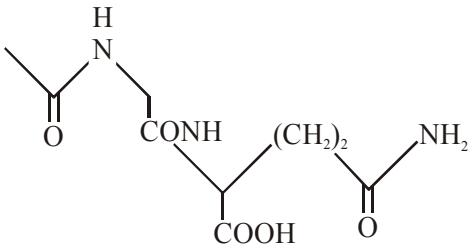
BM0123

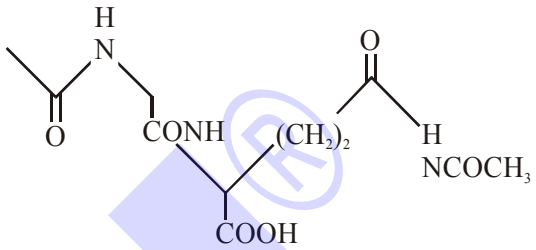
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9. The dipeptide, Gln-Gly, on treatment with CH_3COCl followed by aqueous work work up gives :- **[J-Main-2018]**

(1) 

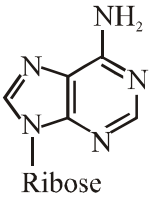
(2) 

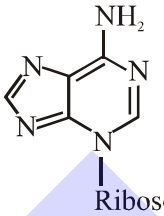
(3) 

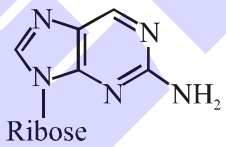
(4) 

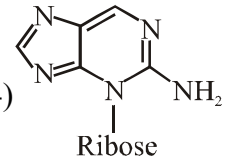
BM0125

10. Which of the following is the correct structure of Adenosine ? **[J-Main-2018]**

(1) 

(2) 

(3) 

(4) 

BM0127

11. Among the following, the incorrect statement is:- **[J-Main-2018]**

(1) Cellulose and amylose has 1, 4-glycosidic linkage.

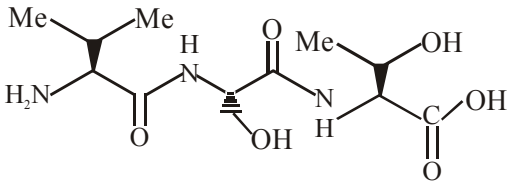
(2) Lactose contains β-D-galactose and β-D-glucose.

(3) Maltose and lactose has 1, 4-glycosidic linkage.

(4) Sucrose and amylose has 1, 2-glycosidic linkage.

BM0128

12. The correct sequence of amino acids present in the tripeptide given below is : **[J-Main-(Jan)-2019]**



(1) Leu - Ser - Thr (2) Thr - Ser- Leu (3) Thr - Ser - Val (4) Val - Ser - Thr

BM0130

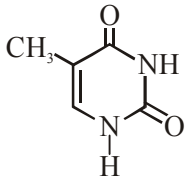
13. Which of the following tests cannot be used for identifying amino acids ? **[J-Main-(Jan)-2019]**

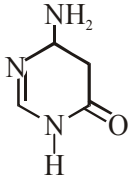
(1) Biuret test (2) Xanthoproteic test (3) Barfoed test (4) Ninhydrin test

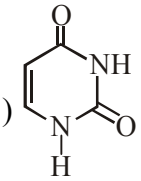
BM0133

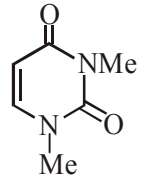
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14. Among the following compound which one is found in RNA ? **[J-Main-(Jan)-2019]**

(1) 

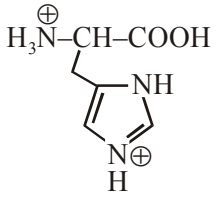
(2) 

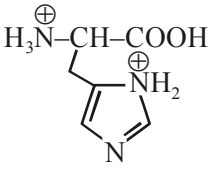
(3) 

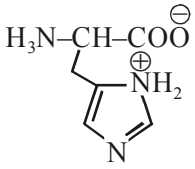
(4) 

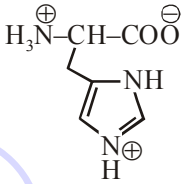
BM0135

15. The correct structure of histidine in a strongly acidic solution (pH=2) is **[J-Main-(Jan)-2019]**

(1) 

(2) 

(3) 

(4) 

BM0137

16. Maltose on treatment with dilute HCI gives : **[J-Main-(April)-2019]**

(1) D-Galactose (2) D-Glucose

(3) D-Glucose and D-Fructose (4) D-Fructose

BM0139

17. Fructose and glucose can be distinguished by : **[J-Main-(April)-2019]**

(1) Fehling's test (2) Barfoed's test (3) Benedict's test (4) Seliwanoff's test

BM0140

18. The peptide that gives positive ceric ammonium nitrate and carbylamine tests is : **[J-Main-(April)-2019]**

(1) Lys-Asp (2) Ser-Lys (3) Gln-Asp (4) Asp-Gln

BM0141

19. Which of the following statements is not true about sucrose? **[J-Main-(April)-2019]**

(1) On hydrolysis, it produces glucose and fructose

(2) The glycosidic linkage is present between C_1 of α-glucose and C_1 of β-fructose

(3) It is also named as invert sugar

(4) It is a non reducing sugar

BM0142

20. Number of stereo centers present in linear and cyclic structures of glucose are respectively : **[J-Main-(April)-2019]**

(1) 4 & 5 (2) 5 & 5

(3) 4 & 4 (4) 5 & 4

BM0143

21. Amylopectin is composed of : **[J-Main-(April)-2019]**

(1) α-D-glucose, C_1–C_4 and C_1–C_6 linkages (2) α-D-glucose, C_1–C_4 and C_2–C_6 linkages

(3) β-D-glucose, C_1–C_4 and C_2–C_6 linkages (4) β-D-Glucose, C_1–C_4 and C_1–C_6 linkages

BM0144

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22. Which of the following statements is not true about RNA ? **[J-Main-(April)-2019]**

(1) It has always double stranded α-helix structure (2) It usually does not replicate

(3) It is present in the nucleus of the cell (4) It controls the synthesis of protein

BM0145

23. Which of the following statements is correct- **[J-Main-(Jan)-2020]**

(1) Gluconic acid can form cyclic (acetal/ hemiacetal) structure

(2) Gluconic acid is a partial oxidation product of glucose

(3) Gluconic acid is obtained by oxidation of glucose with HNO_3

(4) Gluconic acid is a dicarboxylic acid **BM0151**

24. Match the following : **[J-Main-(Jan)-2020]**

(i)	Riboflavin	(a) Beriberi
(ii)	Thiamine	(b) Scurvy
(iii)	Pyridoxine	(c) Cheilosis
(iv)	Ascorbic acid	(d) Convulsions

(1) (i)-(c), (ii)-(a), (iii)-(d), (iv)-(b) (2) (i)-(c), (ii)-(d), (iii)-(a), (iv)-(b)

(3) (i)-(d), (ii)-(b), (iii)-(a), (iv)-(c) (4) (i)-(a), (ii)-(d), (iii)-(c), (iv)-(b)

BM0152

25. Two monomers in maltose are : **[J-Main-(Jan)-2020]**

(1) α-D-glucose and β-D-glucose (2) α-D-glucose and α-D-Fructose

(3) α-D-glucose and α-D-glucose (4) α-D-glucose and α-D-galactose

BM0153

26. A, B and C are three biomolecules. The results of the tests performed on them are given below: **[J-Main-(Jan)-2020]**

	Molisch's Test	Barfoed Test	Biuret Test
A	Positive	Negative	Negative
B	Positive	Positive	Negative
C	Negative	Negative	Positive

A, B and C are respectively :

(1) A = Glucose, B = Fructose, C = Albumin (2) A = Lactose, B = Fructose, C = Alanine

(3) A = Lactose, B = Glucose, C = Alanine (4) A = Lactose, B = Glucose, C = Albumin

BM0154

27. Consider the following reactions : **[J-Main-(Sep)-2020]**

(i) Glucose + ROH $\xrightarrow{\text{dry HCl}}$ Acetal $\xrightarrow[(CH_3CO)_2O]{\text{x eq.of}}$ acetyl derivative

(ii) Glucose $\xrightarrow{Ni/H_2}$ A $\xrightarrow[(CH_3CO)_2O]{\text{y eq.of}}$ acetyl derivative

(iii) Glucose $\xrightarrow[(CH_3CO)_2O]{\text{z eq.of}}$ acetyl derivative

'x', 'y' and 'z' in these reactions are respectively.

(1) 5, 6, & 5 (2) 4, 5 & 5 (3) 5, 4 & 5 (4) 4, 6 & 5

BM0184

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28. The correct observation in the following reactions is : **[J-Main-(Sep)-2020]**

$$\text{Sucrose} \xrightarrow[\text{Cleavage (Hydrolysis)}]{\text{Glycosidic bond}} \text{A} + \text{B} \xrightarrow{\text{Seliwanoff's reagent}} ?$$

(1) Formation of blue colour
(2) Formation of violet colour
(3) Formation of red colour
(4) Gives no colour

BM0185

29. Which of the following will react with $CHCl_3$ + alc. KOH ? **[J-Main-(Sep)-2020]**

(1) Adenine and lysine
(2) Adenine and thymine
(3) Adenine and proline
(4) Thymine and proline

BM0186

30. What are the functional groups present in the structure of maltose ? **[J-Main-(Sep)-2020]**

(1) One ketal and one hemiketal
(2) One acetal and one hemiacetal
(3) Two acetals
(4) One acetal and one ketal

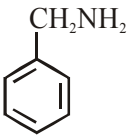
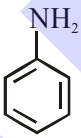
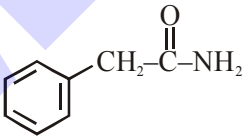
BM0187

31. Which of the following is not an essential amino acid : **[J-Main-(Sep)-2020]**

(1) Valine (2) Leucine (3) Lysine (4) Tyrosine

BM0188

32. Which of the following compounds can be prepared in good yield by Gabriel phthalimide synthesis? **[J-Main-(Sep)-2020]**

(1) 
(2) 
(3) 
(4) $CH_3–CH_2–NHCH_3$

BM0189

33. Which one of the following statements not true ? **[J-Main-(Sep)-2020]**

(1) Lactose contains α-glycosidic linkage between C_1 of galactose and C_4 of glucose.
(2) Lactose ($C_{11}H_{22}O_{11}$) is a disaccharide and it contains 8 hydroxyl groups.
(3) On acid hydrolysis, lactose gives one molecule of D(+)-glucose and one molecule of D(+)-galactose.
(4) Lactose is a reducing sugar and it gives Fehling's test.

BM0190

34. Out of the following, which type of interaction is responsible for the stabilisation of α-helix structure of proteins ? **[J-Main-2021]**

(1) Ionic bonding
(2) Hydrogen bonding
(3) Covalent bonding
(4) vander Waals forces **BM0191**

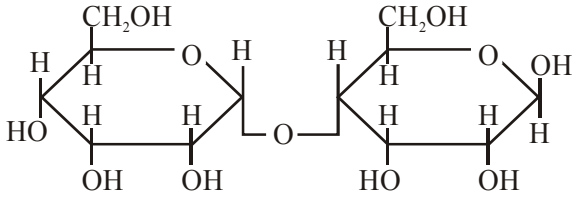
35. Which of the glycosidic linkage between galactose and glucose is present in lactose? **[J-Main-2021]**

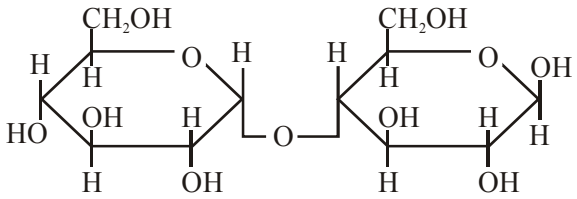
(1) C-1 of galactose and C-4 of glucose
(2) C-1 of glucose and C-6 of galactose
(3) C-1 of glucose and C-4 of galactose
(4) C-1 of galactose and C-6 of glucose

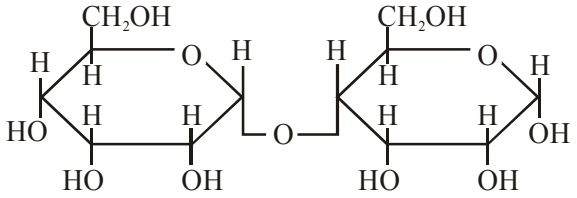
BM0192

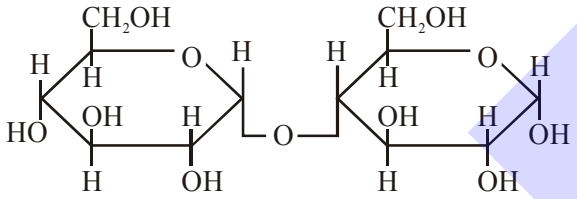
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36. Which of the following is correct structure of α-anomer of maltose ? **[J-Main-2021]**

(1) 

(2) 

(3) 

(4) 

BM0193

37. Which of the following vitamin is helpful in delaying the blood clotting - **[J-Main-2021]**

(1) Vitamin C (2) Vitamin B (3) Vitamin E (4) Vitamin K

BM0194

38. Match List-I with List-II **[J-Main-2021]**

List-I	**List-II**
(a) Sucrose	(i) β-D-Galactose and β-D-Glucose
(b) Lactose	(ii) α-D-Glucose and β-D-Fructose
(c) Maltose	(iii) α-D-Glucose and α-D-Glucose

Choose the correct answer from the options given below :

Options :

(1) (a) → (i), (b) → (iii), (c) → (ii) (2) (a) → (iii), (b) → (i), (c) → (iii)

(3) (a) → (ii), (b) → (i), (c) → (iii) (4) (a) → (iii), (b) → (ii), (c) → (i)

BM0195

39. Which among the following pairs of Vitamins is stored in our body relatively for longer duration ?

(1) Thiamine and Vitamin A (2) Vitamin A and Vitamin D

(3) Thiamine and Ascorbic acid (4) Ascorbic acid and Vitamin D **[J-Main-2021]**

BM0196

40. The secondary structure of protein is stabilised by : **[J-Main-2021]**

(1) Peptide bond (2) glycosidic bond (3) Hydrogen bonding (4) van der Waals forces

BM0197

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41. Fructose is an example of :- **[J-Main-2021]**

(1) Pyranose (2) Ketohexose (3) Aldohexose (4) Heptose

BM0199

42. $C_{12}H_{22}O_{11}+H_2O \xrightarrow{\text{Enzyme A}} C_6H_{12}O_6 + C_6H_{12}O_6$ **[J-Main-2021]**

Sucrose Glucose Fructose

$C_6H_{12}O_6 \xrightarrow{\text{Enzyme B}} 2C_2H_5OH+2CO_2$

Glucose

In the above reactions, the enzyme A and enzyme B respectively are :-

(1) Amylase and Invertase (2) Invertase and Amylase

(3) Invertase and Zymase (4) Zymase and Invertase

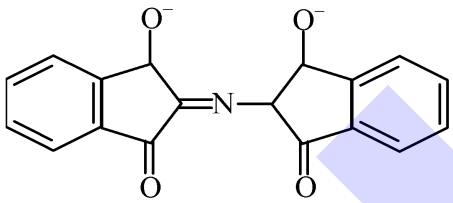
BM0200

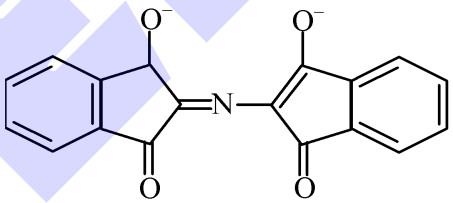
43. A non-reducing sugar "A" hydrolyses to give two reducing mono saccharides. Sugar A is-

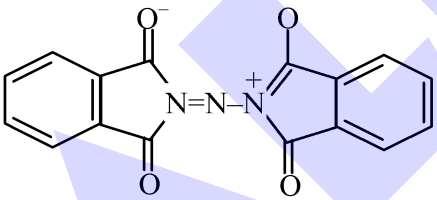
(1) Fructose (2) Galactose (3) Glucose (4) Sucrose **[J-Main-2021]**

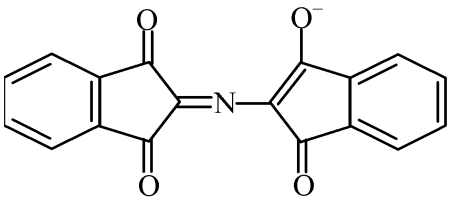
BM0201

44. The correct structure of Rhumann's Purple, the compound formed in the reaction of ninhydrin with proteins is : **[J-Main-2021]**

(1) 

(2) 

(3) 

(4) 

BM0202

45. Identify the incorrect statement from the following **[J-Main-2021]**

(1) Amylose is a branched chain polymer of glucose

(2) Starch is a polymer of α-D glucose

(3) β-Glycosidic linkage makes cellulose polymer

(4) Glycogen is called as animal starch **BM0203**

46. Thiamine and pyridoxine are also known respectively as : **[J-Main-2021]**

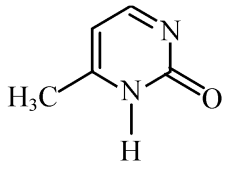
(1) Vitamin B_2 and Vitamin E (2) Vitamin E and Vitamin B_2

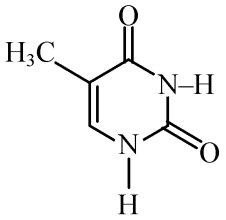
(3) Vitamin B_6 and Vitamin B_2 (4) Vitamin B_1 and Vitamin B_6

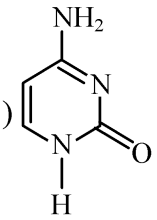
BM0204

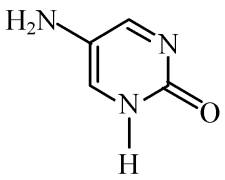
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47. Which one of the following is correct structure for cytosine ? **[J-Main-2021]**

(1) 

(2) 

(3) 

(4) 

BM0205

48. The water soluble protein is : **[J-Main-2021]**

(1) Fibrin (2) Albumin (3) Myosin (4) Collagen

BM0206

49. Compound A gives D-Galactose and D-Glucose on hydrolysis. The compound A is :

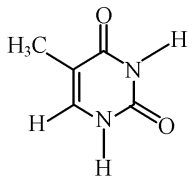
(1) Amylose (2) Sucrose (3) Maltose (4) Lactose **[J-Main-2021]**

BM0207

50. Which one among the following chemical tests is used to distinguish monosaccharide from disaccharide ? **[J-Main-2021]**

(1) Seliwanoff's test (2) Iodine test (3) Barfoed test (4) Tollen's test

BM0208

51. 

(A)

The compound 'A' is a complementary base of _____ in DNA stands. **[J-Main-2021]**

(1) Uracil (2) Guanine (3) Adenine (4) Cytosine

BM0209

52. Given below are two statements : one is labelled as Assertion (A) and other is labelled as Reason (R). **[J-Main-2021]**

Assertion (A) : Sucrose is a disaccharide and a non-reducing sugar.

Reason (R) : Sucrose involves glycosidic linkage between C_1 of β-glucose and C_2 of α-fructose.

Choose the most appropriate answer from the options given below :

(1) Both (A) and (R) are true but (R) is not the true explanation of (A)

(2) (A) is false but (R) is true.

(3) (A) is true but (R) is false

(4) Both (A) and (R) are true and (R) is the true explanation of (A)

BM0210

53. Hydrolysis of sucrose gives : **[J-Main-2021]**

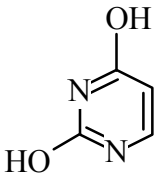
(1) α-D-(-)-Glucose and β-D-(-)-Fructose (2) α-D-(+)-Glucose and α-D-(-)-Fructose

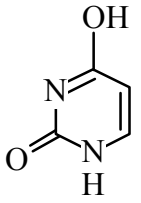
(3) α-D-(-)-Glucose and α-D-(+)-Fructose (4) α-D-(+)-Glucose and β-D-(-)-Fructose

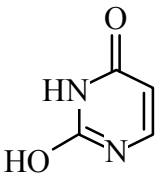
BM0211

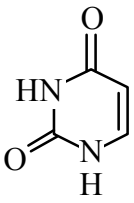
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54. Out of following isomeric forms of uracil, which one is present in RNA ? **[J-Main-2021]**

(1) 

(2) 

(3) 

(4) 

BM0212

55. Which of the following is NOT an example of fibrous protein ? **[J-Main-2021]**

(1) Keratin (2) Albumin (3) Collagen (4) Myosin

BM0213

56. Which one of the following compounds contains β-C_1-C_4 glycosidic linkage ? **[J-Main-2021]**

(1) Lactose (2) Sucrose (3) Maltose (4) Amylose

BM0214

57. Glycosidic linkage between C_1 of α-glucose and C_2 of β-fructose is found in **[J-Main-2022]**

(1) maltose (2) sucrose (3) lactose (4) amylose

BM0233

58. During the denaturation of proteins, which of these structures will remain intact ? **[J-Main-2022]**

(1) Primary (2) Secondary (3) Tertiary (4) Quaternary

BM0234

59. Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Amylose is insoluble in water. **[J-Main-2022]**

Reason R : Amylose is a long linear molecule with more than 200 glucose units.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both A and R are correct and R is the correct explanation of A.

(2) Both A and R are correct and R is NOT the correct explanation of A.

(3) A is correct but R is not correct.

(4) A is not correct but R is correct. **BM0235**

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60. Animal starch is the other name of : **[J-Main-2022]**

(1) amylose (2) maltose (3) glycogen (4) amylopectin

BM0236

61. A sugar 'X' dehydrates very slowly under acidic condition to give furfural which on further reaction with resorcinol gives the coloured product after sometime. Sugar 'X' is **[J-Main-2022]**

(1) Aldopentose (2) Aldotetrose (3) Oxalic acid (4) Ketotetrose

BM0237

62. Match List-I with List-II **[J-Main-2022]**

List-I	List-II
(A) Glucose + HI	(I) Gluconic acid
(B) Glucose + water	(II) Glucose pentacetate
(C) Glucose + acetic anhydride	(III) Saccharic acid
(D) Glucose + HNO_3	(IV) Hexane

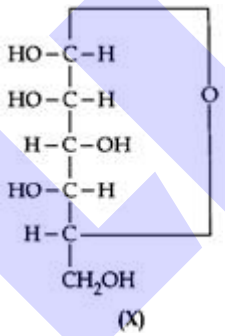
Choose the correct answer from the options given below:

(1) (A)-(IV), (B)-(I), (C)-(II), (D)-(III) (2) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

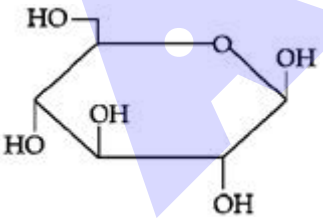
(3) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) (4) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

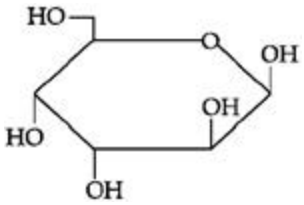
BM0238

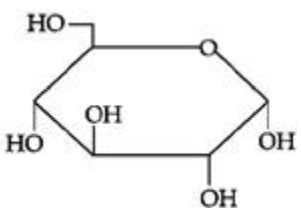
63. For the below given cyclic hemiacetal (X), the correct pyranose structure is : **[J-Main-2022]**

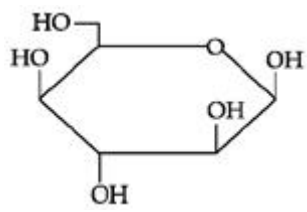


(X)

(1) 

(2) 

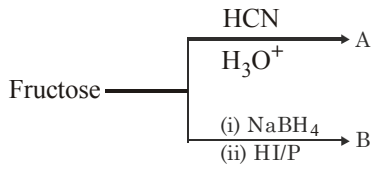
(3) 

(4) 

BM0239

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64. The formulas of A and B for the following reaction sequence are **[J-Main-2022]**



(1) A = $C_7H_{14}O_8$, B = C_6H_{14}
(2) A = $C_7H_{13}O_7$, B = $C_7H_{14}O$
(3) A = $C_7H_{12}O_8$, B = C_6H_{14}
(4) A = $C_7H_{14}O_8$, B = $C_6H_{14}O_6$

BM0240

INTEGER TYPE

65. The number of $\rangle C = O$ groups present in a tripeptide Asp – Glu – Lys is _____. **[J-Main-(Sep)-2020]**

BM0215

66. The number of chiral centres present in threonine is _____. **[J-Main-(Sep)-2020]**

BM0216

67. The number of chiral carbon(s) present in peptide, Ile-Arg-Pro, is _____. **[J-Main-(Sep)-2020]**

BM0217

68. The number of chiral carbons present in sucrose is_____ . **[J-Main-(Sep)-2020]**

BM0218

69. A peptide synthesized by the reactions of one molecule each of Glycine, Leucine, Aspartic acid and Histidine will have _____ peptide linkages. **[J-Main-2021]**

BM0219

70. The total number of negative charge in the tetrapeptide, Gly-Glu-Asp-Tyr, at pH 12.5 will be _____. **[J-Main-2021]**

BM0220

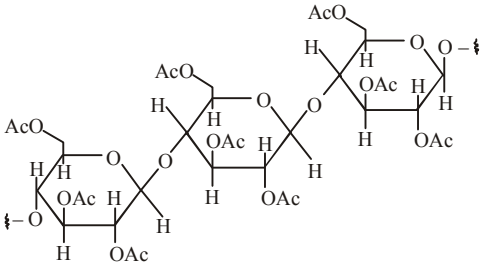
71. In a linear tetrapeptide (Constituted with different amino acids), (number of amino acids) - (number of peptide bonds) is_____. **[J-Main-2022]**

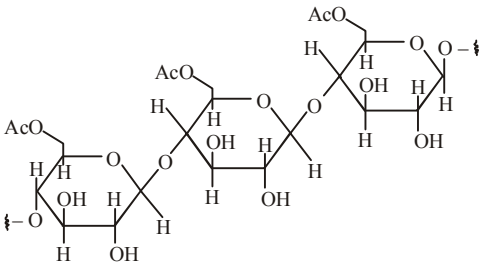
BM0241

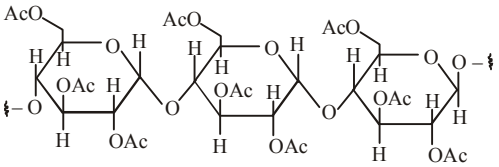
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\S-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che\Every day\Eng\3-Biomole\01_Th & Ex.p65

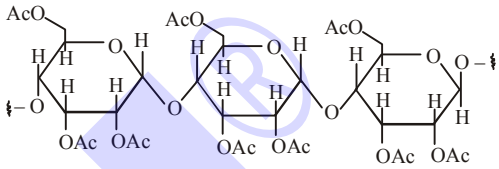
EXERCISE # J-ADVANCED_(OBJECTIVE)

1. Cellulose upon acetylation with excess acetic anhydride / H_2SO_4 (catalytic) gives cellulose triacetate whose structure is **[JEE 2008]**

(A) 

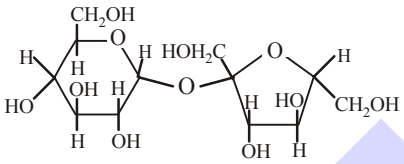
(B) 

(C) 

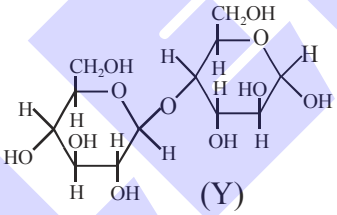
(D) 

BM0158

2. The correct statement(s) about the following sugars X and Y is(are) **[JEE 2009]**



(X)



(Y)

(A) X is a reducing sugar and Y is a non-reducing sugar

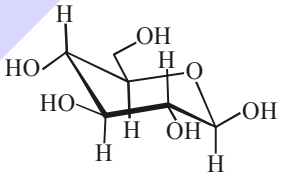
(B) X is a non-reducing sugar and Y is a reducing sugar

(C) The glucosidic linkages in X and Y are α and β , respectively.

(D) The glucosidic linkages in X and Y are β and α , respectively

BM0160

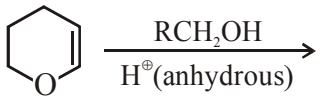
3. The following carbohydrate is **[JEE 2011]**



(A) a ketohexose (B) an aldohexose (C) an α-furanose (D) an α-pyranose

BM0162

4. The major product of the following reaction is **[JEE 2011]**

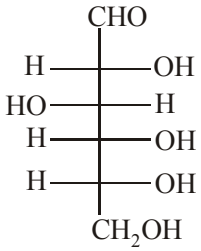


(A) a hemiacetal (B) an acetal (C) an ether (D) an ester

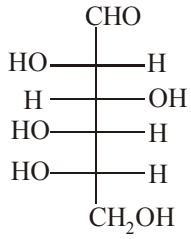
BM0163

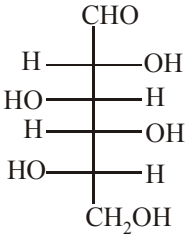
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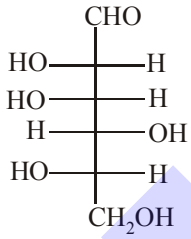
5. The structure of D-(+)-glucose is **[IIT 2011]**

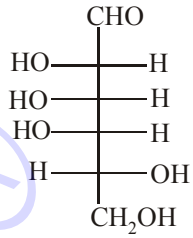


The structure of L(–)-glucose is

(A) 

(B) 

(C) 

(D) 

BM0166

6. For 'invert sugar', the correct statement(s) is (are) : **[JEE 2016]**

(Given : specific rotations of (+)-sucrose, (+)-maltose, L-(–)-glucose and L-(+)-fructose in aqueous solution are + 66°, +140°, –52° and + 92°, respectively)

(A) 'invert sugar' is prepared by acid catalyzed hydrolysis of maltose

(B) 'invert sugar' is an equimolar mixture of D-(+) glucose and D-(–)-fructose

(C) specific rotation of 'invert surgar' is –20°

(D) on reaction with Br_2 water, 'invert sugar' forms saccharic acid as one of the products

BM0167

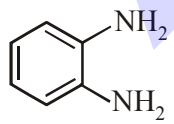
Comprehension : Q.No. 7 to 8

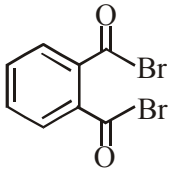
Treatment of compound **O** with $KMnO_4 / H^+$ gave **P**, which on heating with ammonia gave **Q**. The compound **Q** on treatment with Br_2 / NaOH produced **R**. On strong heating, **Q** gave **S**, which on further treatmenet with ethyl 2-bromopropanoate in the presence of KOH following by acidification, gave a compound **T**.

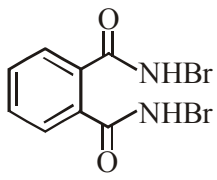


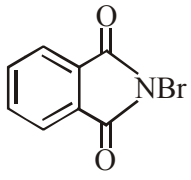
(O)

7. The compound **R** is : **[JEE 2016]**

(A) 

(B) 

(C) 

(D) 

BM0168

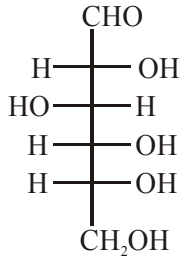
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8. The compound **T** is : **[JEE 2016]**

(A) Glycine (B) Alanine (C) Valine (D) Serine

BM0169

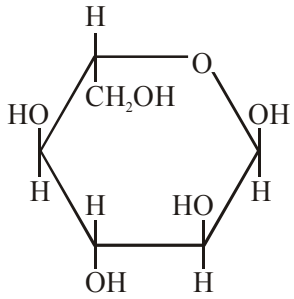
9. The Fischer presentation of D-glucose is given below. **[JEE 2018]**



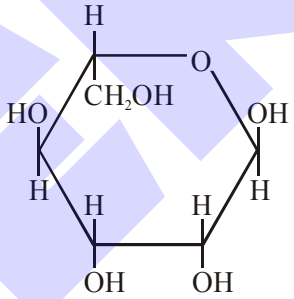
D-glucose

The correct structure(s) of β-L-glucopyranose is (are) :-

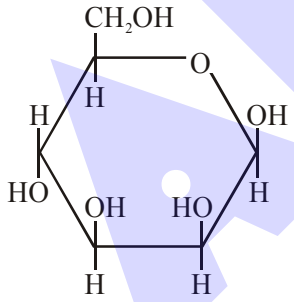
(A)



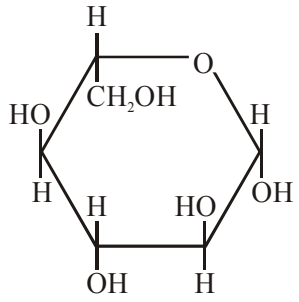
(B)



(C)



(D)



BM0170

10. Which of the following statement(s) is(are) true ? **[JEE 2019]**

(A) Oxidation of glucose with bromine water gives glutamic acid

(B) The two six-membered cyclic hemiacetal forms of D-(+)-glucose ard called anomers

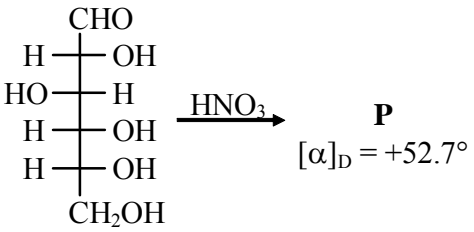
(C) Hydrolysis of sucrose gives dextrorotatory glucose and laevorotatory fructose

(D) Monosaccharides **cannot** be hydrolysed to give polyhydroxy aldehydes and ketones

BM0171

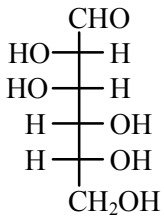
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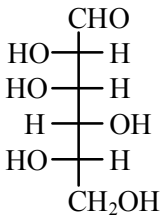
11. Given **[JEE 2021]**

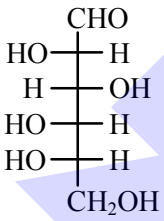


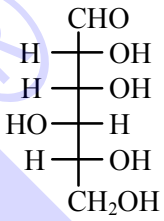
D-Glucose

The compound(s), which on reaction with HNO_3 will give the product having degree of rotation, $[\alpha]_D$ = -52.7° is (are)

(A) 

(B) 

(C) 

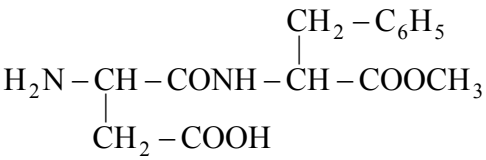
(D) 

BM0221

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EXERCISE # J-ADVANCED_(SUBJECTIVE)

1. Aspartame, an artifical sweetener, is a peptide and has the following structure: **[JEE 2001]**



(i) Identify the four functional groups

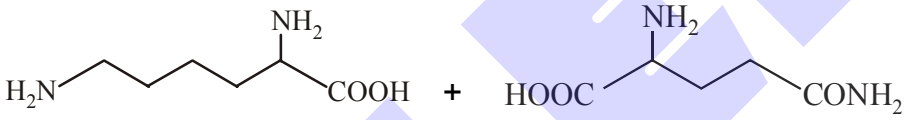
(ii) Write the zwitterionic structure

(iii) Write the structures of the amino acids obtained from the hydrolysis of aspartame

(iv) Which of the two amino acids is more hydrophobic ?

BM0174

2. Following two amino acids lysine and glutamine form dipeptide linkage. What are two possible dipeptides ? **[JEE 2003]**

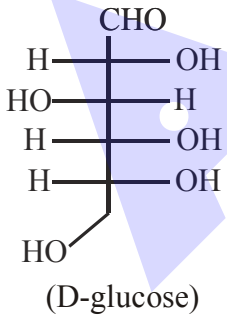


BM0175

3. The structure of D-Glucose is as follows- **[JEE 2004]**

(a) Draw the structure of L -Glucose

(b) Give the reaction of L - Glucose with Tollens reagent.



(D-glucose)

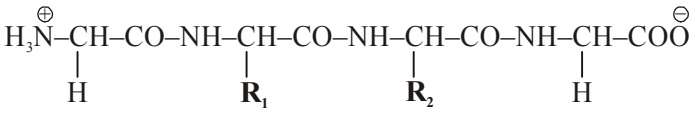
BM0176

4. A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt. 75), alanine and phenylalanine. Glycine contributes 47.0% to the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is **[JEE 2011]**

BM0178

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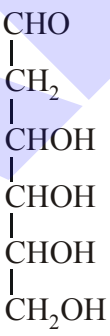
5. The substitutes $\mathbf{R_1}$ and $\mathbf{R_2}$ for nine peptides are listed in the table given below. How many of these peptides are positively charged at pH = 7.0 ? **[JEE 2012]**



Peptide	$\mathbf{R_1}$	$\mathbf{R_2}$
I	H	H
II	H	CH_3
III	CH_2COOH	H
IV	CH_2CONH_2	$(CH_2)_4NH_2$
V	CH_2CONH_2	CH_2CONH_2
VI	$(CH_2)_4NH_2$	$(CH_2)_4NH_2$
VII	CH_2COOH	CH_2CONH_2
VIII	CH_2OH	$(CH_2)_4NH_2$
IX	$(CH_2)_4NH_2$	CH_3

BM0179

6. When the following aldohexose exists in its d-configuration , the total number of stereoisomers in its pyranose form is - **[JEE 2012]**



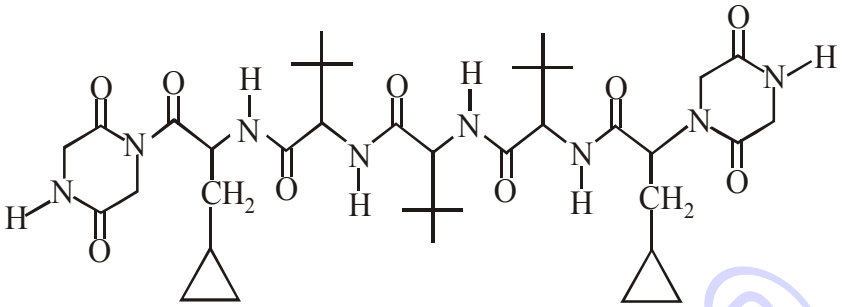
BM0180

7. A tetrapeptide has –COOH group on alanine. This produces glycine (Gly), valine (Val), phenyl alanine (Phe) and alanine (Ala), on complete hydrolysis. For this tetrapeptide, the number of possible sequences (Primary structures) with $-NH_2$ group attached to a chiral center is **[JEE 2013]**

BM0181

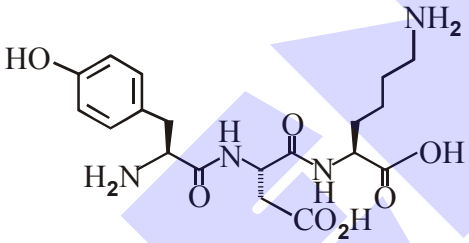
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8. The total number of ***distinct naturally occurring amino acids*** obtained by complete acidic hydrolysis of the peptide shown below is : **[JEE 2014]**



BM0183

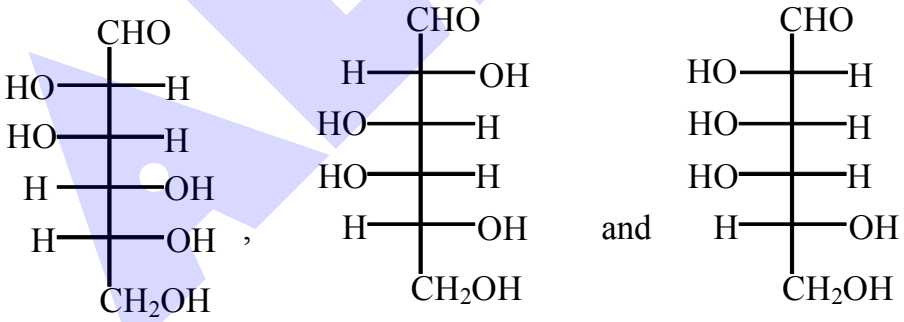
9. The structure of a peptide is given below. **[JEE 2020]**

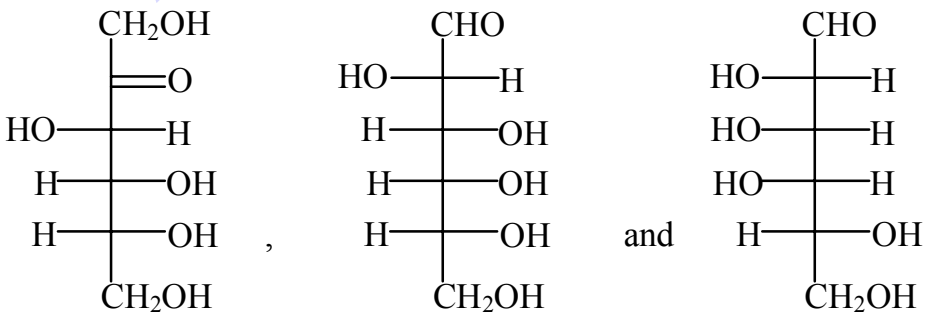


If the absolute values of the net charge of the peptide at pH = 2, pH = 6, and pH = 11 are $|z_1|$, $|z_2|$ and $|z_3|$, respectively, then what is $|z_1| + |z_2| + |z_3|$?

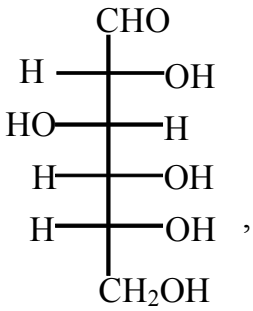
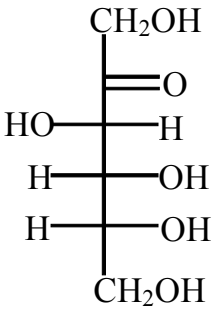
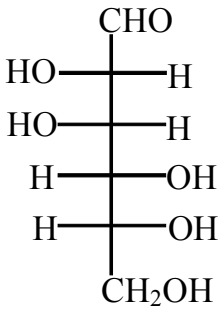
BM0222

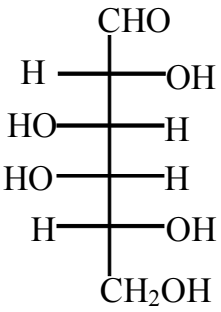
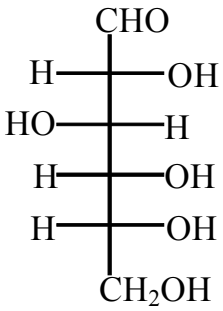
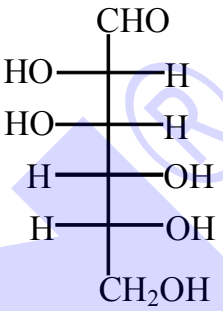
10. Treatment of D- glucose with aqueous NaOH results in a mixture of monosaccharides, which are **[JEE 2022]**

(A) 

(B) 

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(C)  ,  and 

(D)  ,  and 

BM0242

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ANSWER KEY

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	C	B	B	C	C	A	A	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	B	B	D	A	A	C	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	D	C	A	A,B,C,D	A,B,D	A,B,D	B,C	D
Que.	31									
Ans.	A → P, Q, S ; B → S ; C → R, S ; D → Q,R, S									

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,D	A,B,C	C	B	C	D	C	B	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	A	A	B	C	A	B	A	D
Que.	21	22	23	24						
Ans.	B	C	B	D						

EXERCISE # S-I

Que.	1	2	3	4	5	6	7
Ans.	A	A	A	A	D	B	A → S ; B → Q ; C → P ; D → R
Que.	8				9		
Ans.	A → Q,S ; B → Q,R,S ; C → P, R, T, U ; D → P,S,T,U				A → P,Q,R ; B → P; C → P, S ; D → P,Q,R		

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10.

	(i)	(ii)	(iii)
(a)	α, 1, 4	β, 1, 4	α, 1, 2
(b)	Maltose	Lactose	Sucrose
(c)	α-D-Glucose	α-D-Glucose	α-D-Glucose
	& β-D-Glucose	& β-D-Glucose	& β-D-Fructose
	Finally all undercoes mutarotation in aq. solution		
(d)	Reducing sugar	Reducing sugar	Non-Reducing sugar
(e)	4	4	3
(f)	Yes	Yes	No
(g)	8	8	8

Que.	11	12	13	
Ans.	**(i)** (a) 8 (b) 8, **(ii)** (a) 4 (b) 4, **(iii)** (a) 4 (b) 4	32	3.25	

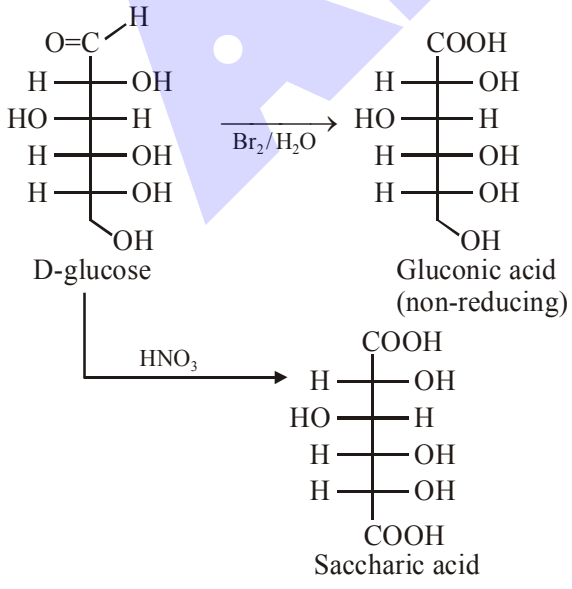
14. p-amino benzoic acid is less acidic and less basic so it does not exist in zwitter ion (dipolar ion)

15. **(a)** = 5.95, **(b)** = 5.70, **(c)** = 6.00, **(d)** = 3.00, **(e)** = 9.75

EXERCISE # J-MAIN

1. (2)	**2. (2)**	**3. (3)**	**4. (3)**	**5. (3)**	**6. (4)**
7. (1)	**8. (4)**	**9. (1)**	**10. (1)**	**11. (4)**	**12. (4)**
13. (3)	**14. (3)**	**15. (1)**	**16. (2)**	**17. (4)**	**18. (2)**
19. (2)	**20. (1)**	**21. (1)**	**22. (1)**		
23. (2)					

Sol.



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24. (1)

Sol. (i) Riboflavin ⟶ (c) Cheilosis

(ii) Thiamine ⟶ (a) Beriberi

(iii) Pyridoxin ⟶ (d) Convulsions

(iv) Ascorbic acid ⟶ (b) Scurvy

25. (3)

Sol. Two monomers in maltose are α-D-glucose & α-D-glucose.

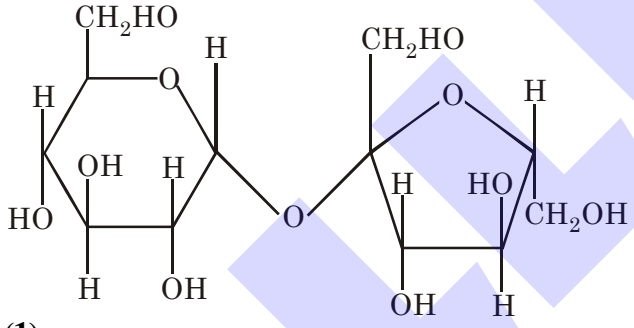
26. (4)

Sol. Alanine does not show Biuret test because Biuret test is used for deduction of peptide linkage & alanine is amino acid.
Albumine is protein so have paptide linkage so it gives positive Biuret test.
Positive Barfoed test is shown by monosaccharide but not disaccharide. Positive Molisch's test is shown by glucose.

27. (4)	**28. (3)**	**29. (1)**	**30. (2)**	**31. (4)**	**32. (1)**
33. (1)	**34. (2)**	**35. (1)**	**36. (4)**	**37. (3)**	**38. (3)**
39. (2)	**40. (3)**	**41. (2)**	**42. (3)**	**43. (4)**	**44. (4)**
45. (1)	**46. (4)**	**47. (3)**	**48. (2)**	**49. (4)**	**50. (3)**
51. (3)	**52. (3)**	**53. (4)**	**54. (4)**	**55. (2)**	**56. (1)**

57. (2)

Sol. Theoretical



58. (1)

Sol. Primary structure remains intact during denaturation of proteins

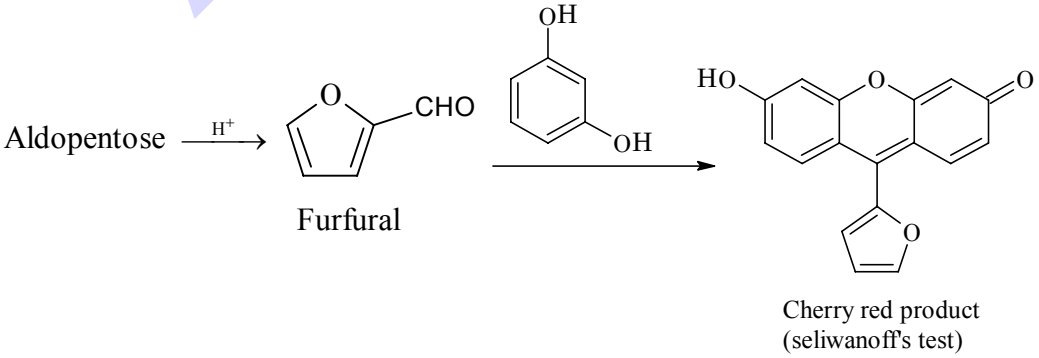
59. (4)

Sol. Amylose is water soluble.

60. (3)

Sol. Glycogen

61. (1)



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62. (1)

Sol. (1) Glucose $\xrightarrow{HI}$ n-hexane

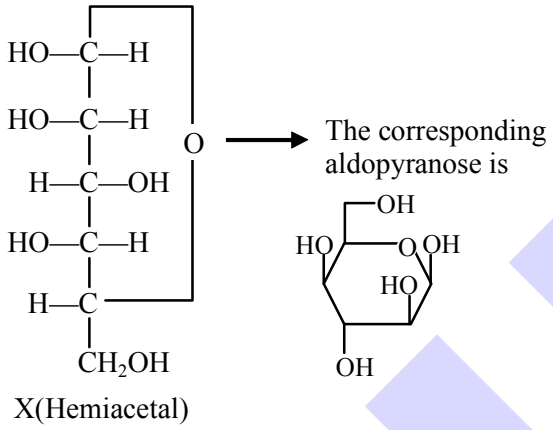
(2) Glucose $\xrightarrow[H_2O]{Br_2}$ $\begin{matrix}COOH\\|\\(CHOH)_4\\|\\CH_2OH\end{matrix}$ Gluconic acid

(3) Glucose $\xrightarrow[\text{anhydride}]{\text{5 acetic}}$ Glucose pentacetate

(4) Glucose $\xrightarrow{HNO_3}$ $\begin{matrix}COOH\\|\\(CHOH)_4\\|\\COOH\end{matrix}$ Saccharic acid

63. (4)

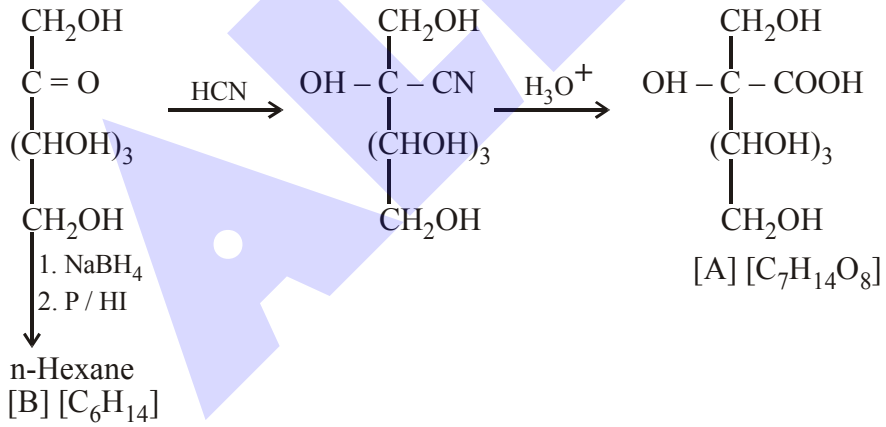
Sol. Correct pyranose structure is



X(Hemiacetal)

64. (1)

Sol.



65. (5) **66. (2)** **67. (4)** **68. (9)**

69. (3) **70. (4)**

71. (1)

Sol. In Tetrapeptide,

No. of Amino Acids = 4

No. of Peptide bonds = 3

Hence = 1

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EXERCISE # J-ADVANCED_(OBJECTIVE)

1. (A) **2. (B,C)** **3. (B)** **4. (B)** **5. (A)**

6. (B,C) **7. (A)** **8. (B)** **9. .(D)** **10. (B,C,D)**

11. (C,D)

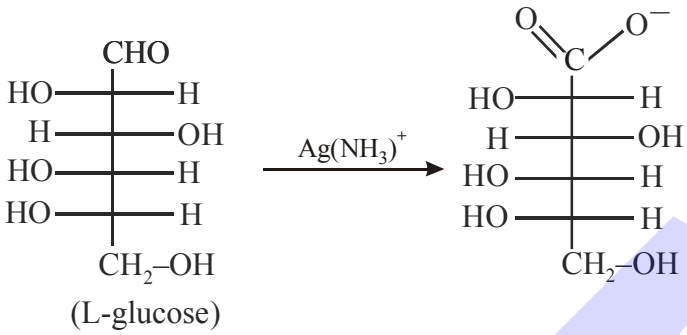
EXERCISE # J-ADVANCED_(SUBJECTIVE)

1. (i) Amine, carboxylic acid, Amide, Ester

2. (2)

Lys–Gln, Gln–Lys

3.



4. (6) 5. (4) **6. (8)** **7. (4)**

8. (1) 9. (5) **10. (C)**

ALLEN®

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